

Correcting photometric observations for cloud extinction using thermal infrared observations

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AAS 247 Winter Meeting, January 4–8 2026, Phoenix, Arizona



Preprint: [arXiv:2511.00182](https://arxiv.org/abs/2511.00182)
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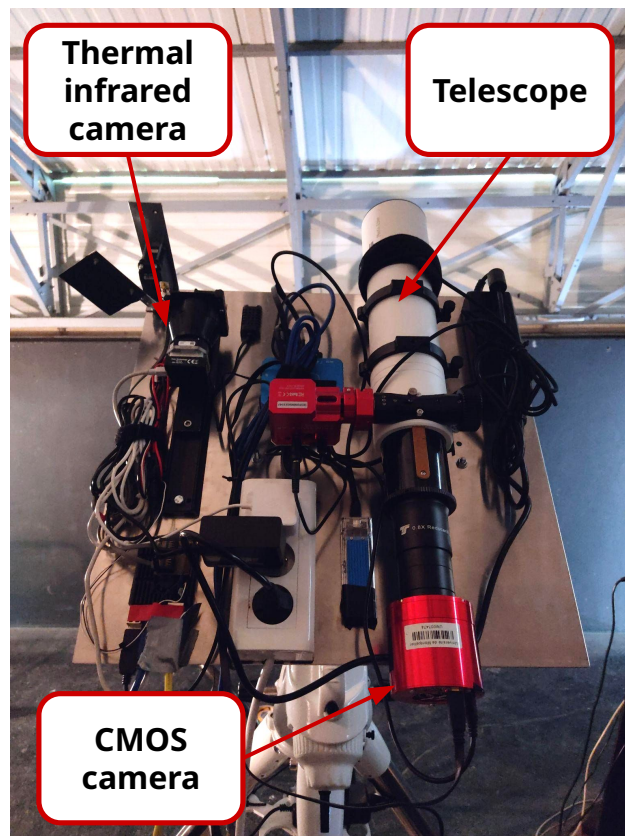
Core principles

- Clouds (water droplets & ice crystals) **emit thermally in the LWIR (8–13 μm)** at ~200–300 K (Heymsfield et al., 2017)
- **Visible and IR cloud optical depths scale proportionally** (DeSlover et al., 1999)
- By measuring an atmosphere down-welling radiance excess ΔL in a star line-of-sight, it is possible to infer the loss of magnitude Δm as it is proportional to the cloud optical depth:

$$\Delta m \propto \tau_{\text{cloud}} \propto \Delta L$$

Key ingredients

- An **infrared thermal camera** that underwent radiometric calibration
- **Local atmospheric monitoring** $\Rightarrow$ H_2O , temperature, pressure, ozone, etc.
- Knowledge of the **instrumental transmission**
- **Sample of calibrators** with known stellar parameters for synthetic photometry $\Rightarrow$ Gaia DR3 catalogs
- Lookup tables of **synthetic stellar spectra collection** $\Rightarrow$ AMBRE (P. de Laverny et al., 2012)
- **Instrumental photometry** of calibrators $\Rightarrow$ Source Extractor (Bertin & Arnouts 1996)
- **Atmospheric radiative transfer modeling** $\Rightarrow$ libRadtran (C. Emde et al., 2016)

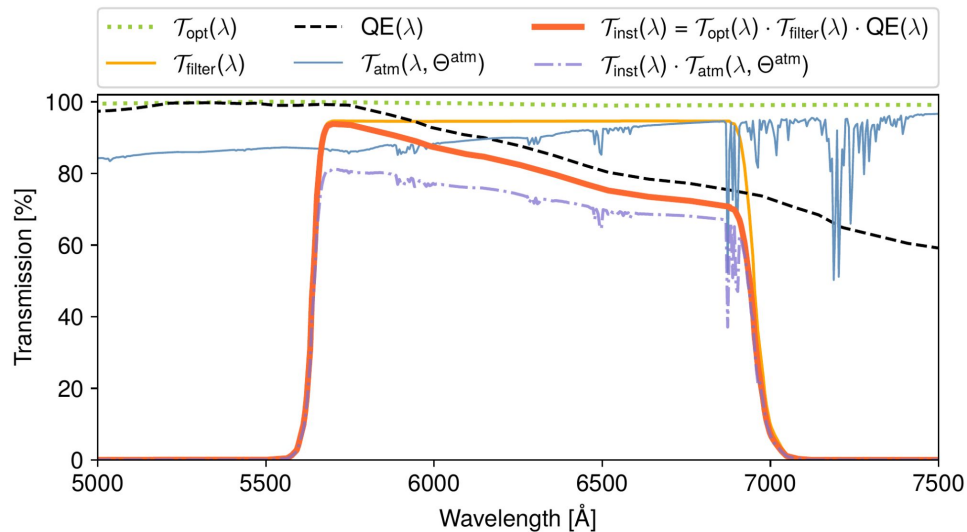


Transmission model

$$\mathcal{T}(\lambda) = \mathcal{T}^{\text{atm}}(\lambda, \Theta_i^{\text{atm}}) \times \mathcal{T}^{\text{inst}}(\lambda)$$

$$\mathcal{T}^{\text{atm}}(\lambda, \Theta_i^{\text{atm}}) = \mathcal{T}^{\text{chromatic}}(\lambda, \Theta_i^{\text{atm}}) \times \mathcal{T}^{\text{gray}}(t_i, \text{alt}_i, \text{az}_i)$$

$$\mathcal{T}^{\text{inst}}(\lambda) = \mathcal{T}^{\text{opt}}(\lambda) \times \mathcal{T}^{\text{filter}}(\lambda) \times QE(\lambda)$$



i = image epoch

Chromatic atmosphere component removal

- Instrumental magnitude: $m_{b,s,i,k}^{\text{inst}} = -2.5 \times \log_{10} \left(\frac{ADU_{b,s,i,k}}{t_{\text{exp}}} \right) + \frac{\Delta_{b,i,k}}{\text{Aperture correction}}$

- Chromatic atmosphere component:

$$\mathcal{C}_{s,i} = 2.5 \log_{10} \left(\underbrace{\int_0^\infty \mathcal{S}_s(\lambda, T_{\text{eff}}, \log g, \text{Fe}/\text{H}) \times \mathcal{T}_b^{\text{inst}}(\lambda) \times [1 - \mathcal{T}^{\text{chromatic}}(\lambda, \Theta_i)] \times \frac{\lambda}{hc} \times d\lambda}_{\substack{\text{SED with Gaia DR3} \\ \text{+ lookup tables}}} \times \underbrace{\frac{\lambda}{hc} \times d\lambda}_{\text{Simulated}} \right)$$

- Top-of-(chromatic)-atmosphere instrumental magnitude:

$$m_{s,i}^{\text{TOA, inst}} = m_{s,i}^{\text{inst}} - \mathcal{C}_{s,i}$$

b = band
s = star
i = image epoch
k = aperture radius

Gray extinction model optimization

- Gray extinction: $\Delta m_{s,i} \equiv m_{s,i}^{\text{TOA, inst}} - \underbrace{m_s^{\text{TOA, ref}}}_{\text{Reference magnitude of the calibrator}}$

- Fit the gray extinction estimates with the radiance excess on calibrators:

$$\hat{\Delta} m_{s,i} = -\frac{1}{B_i} \times \log \left(\frac{A_i + C_i - \Delta L_{s,i}}{A_i} \right)$$

Physical model (high-extinction range > 1 mag)

$$\hat{\Delta} m_{s,i} = A_i \times \Delta L_{s,i} + B_i$$

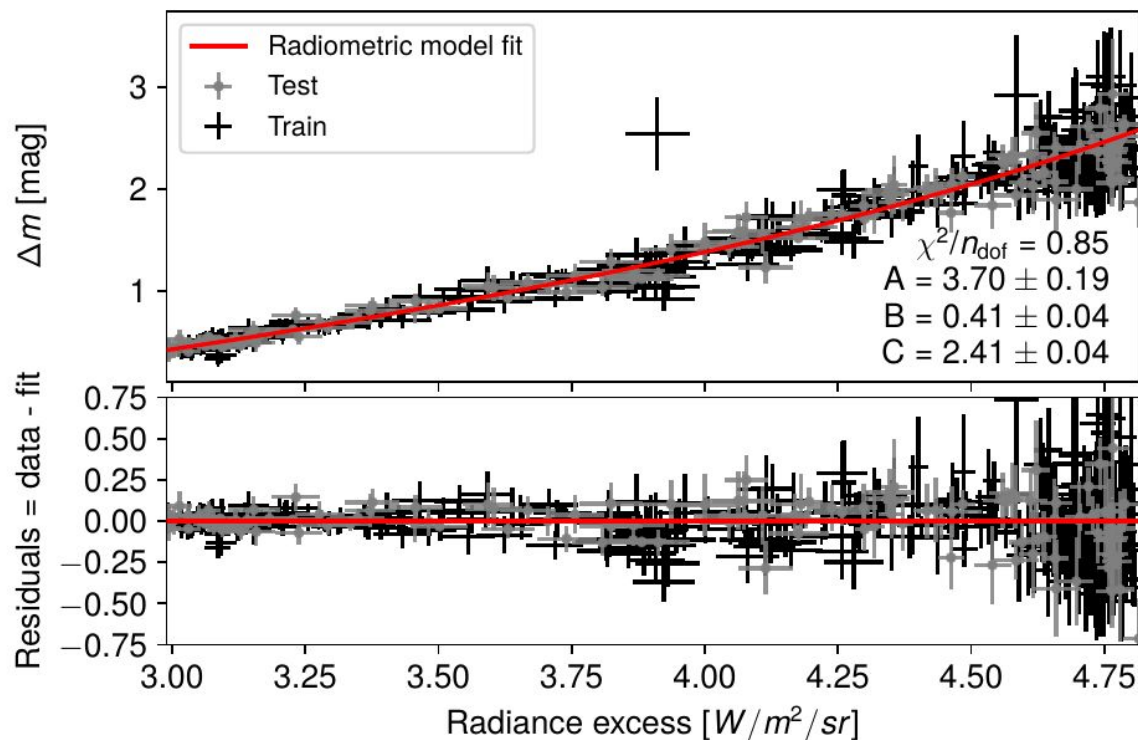
Linear model (low-extinction range $\ll$ 1 mag)

- Residuals to be minimized: $R_{s,i} = \Delta m_{s,i} - \hat{\Delta} m_{s,i}$

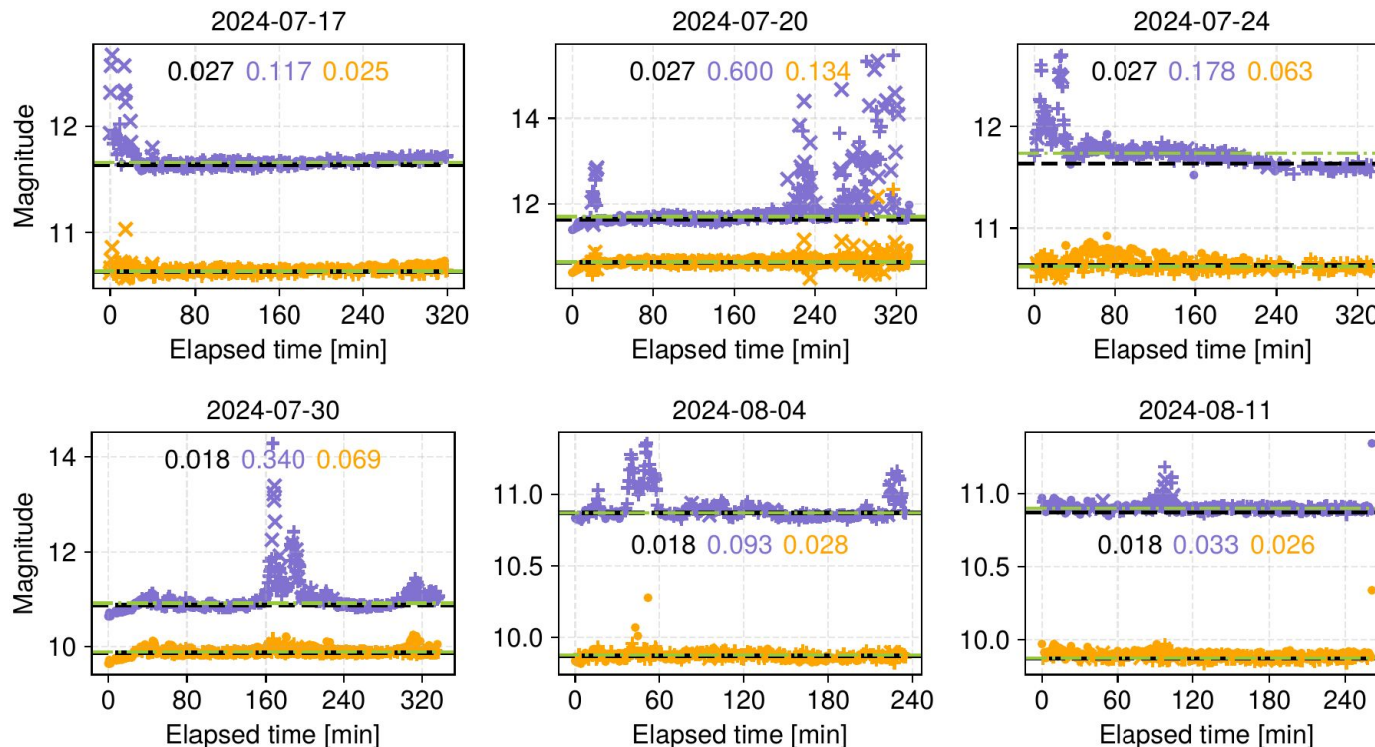
- Corrected magnitudes: $m_{s,i}^{\text{TOA, corr}} = m_{s,i}^{\text{TOA, inst}} - \hat{\Delta} m_{s,i}$

b = band
s = star
i = image epoch
k = aperture radius

Diagnostic plot after fit



Impact on light curves of a test star



Conclusion and next steps

- Proof-of-concept successfully demonstrated
- Improve photometric data quality for variable and poor conditions:
 - Recover data down to 0.025 mag precision for sequences affected by as much as 3.0 magnitudes of extinction.
 - For the most affected exposures, the mean absolute error on photometry improves from 0.64 to 0.11 mag.
- Need to evaluate the “gray” nature of extinction by observing in multiple bands (e.g., *ugrizy* with the StarDICE experiment telescope)

Backup slides

Context

- Ground-based time-domain astronomy surveys such as the Vera C. Rubin Observatory's *Legacy Survey of Space and Time* require photometric calibration that is both **long-term stable and spatially uniform at the sub-percent level, even during non-photometric conditions** (Burke D. L., et al., 2013, ApJ, 147, 19) $\Rightarrow$ given the large FOV (about 9.6 deg) of the Rubin LSST camera, it is unrealistic to assume that cloud structures are uniform across the entire FOV.
- **The key issue is no longer whether the night is photometric, but rather determining the extent and structure of cloud(s).**
- Hazenberg et al. 2019 and Burke D. L., et al., 2013 proposed an approach which consists in **gathering independent measurements of atmospheric downwelling radiance using an infrared thermal camera sensitive to the 8-13 μm range.**

Context

- The *StarDICE* experiment aims at improving the photometric calibration of CALSPEC standard stars by establishing a metrology chain linking NIST laboratory standards to astrophysical fluxes with 1 mmag accuracy in the *griz* bands ⇒ **controlling variable atmospheric effects is essential.**
- *StarDICE pathfinder* experiment demonstrated that even **small amounts of extinction variations caused by passing undetected clouds, alters the measurement and drastically reduces the usable statistics** (Hazenbergh 2019, PhD manuscript).

The StarDICE experiment

Aims at improving the photometric calibration of CALSPEC standard stars to the mmag level, by monitoring instrument throughput with a LED-based artificial star source, calibrated on NIST photodiodes.

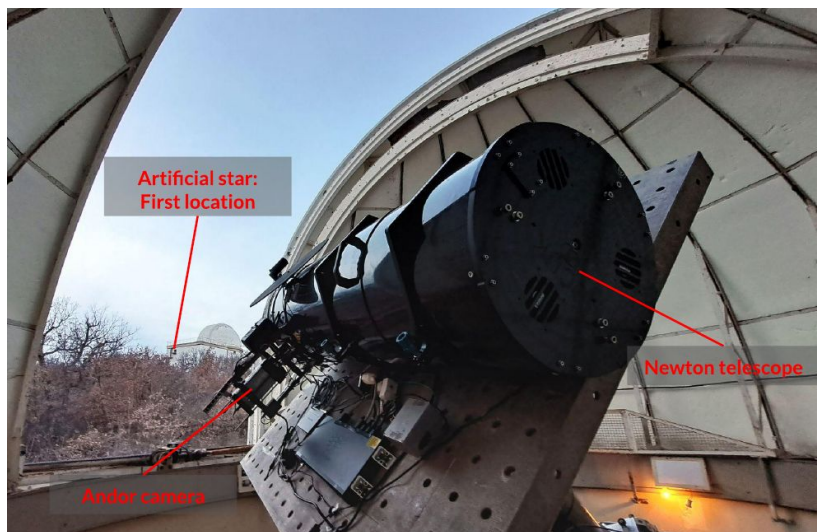


Figure 1 – StarDICE telescope on its equatorial mount at Observatoire de Haute-Provence, France.

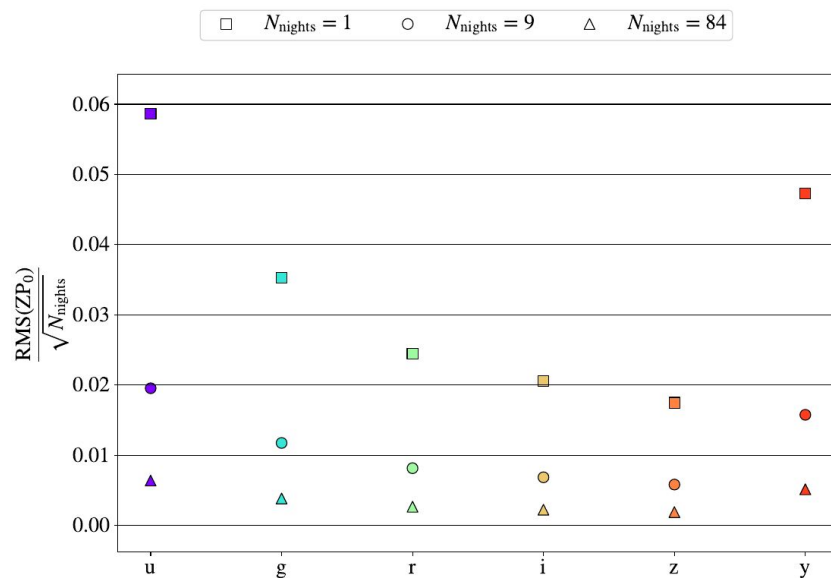
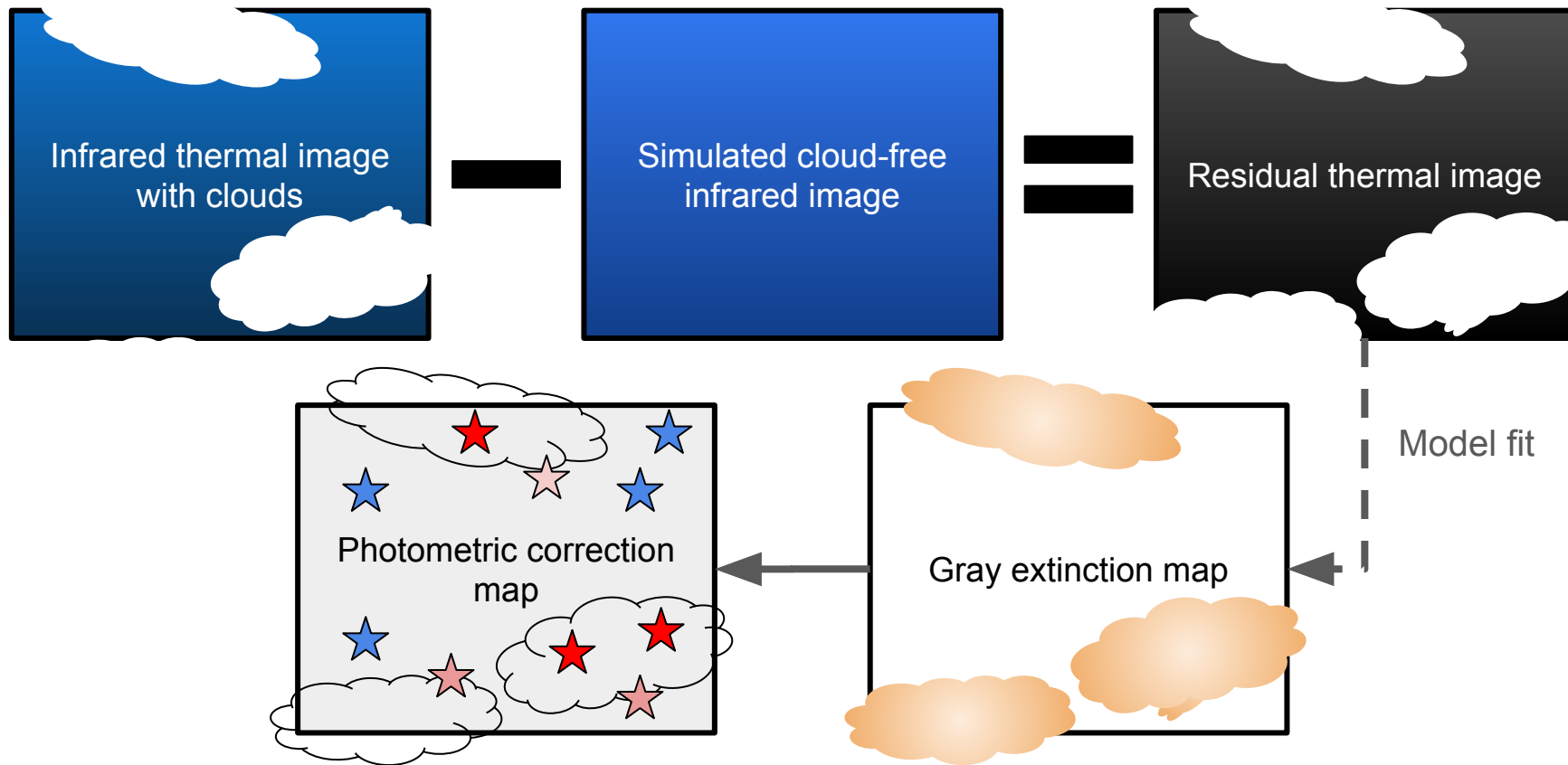
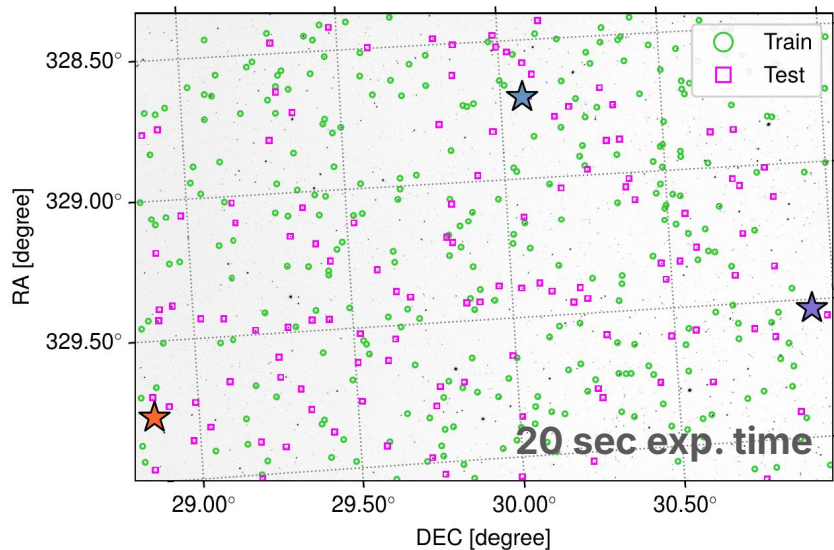


Figure 2 – RMS of zeropoint fitted for the 9 photometric nights of observations of G191B2B and for each StarDICE band, normalized by the square root of the number of nights.

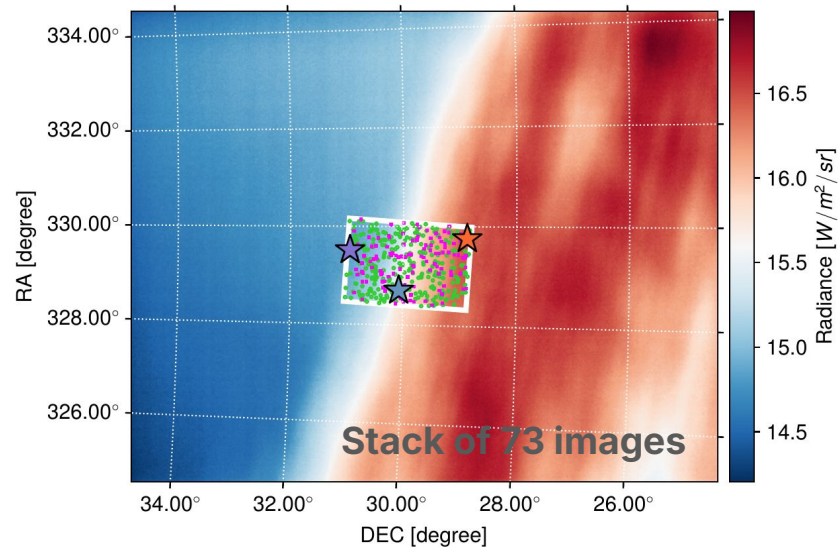


Data overview

CMOS FOV

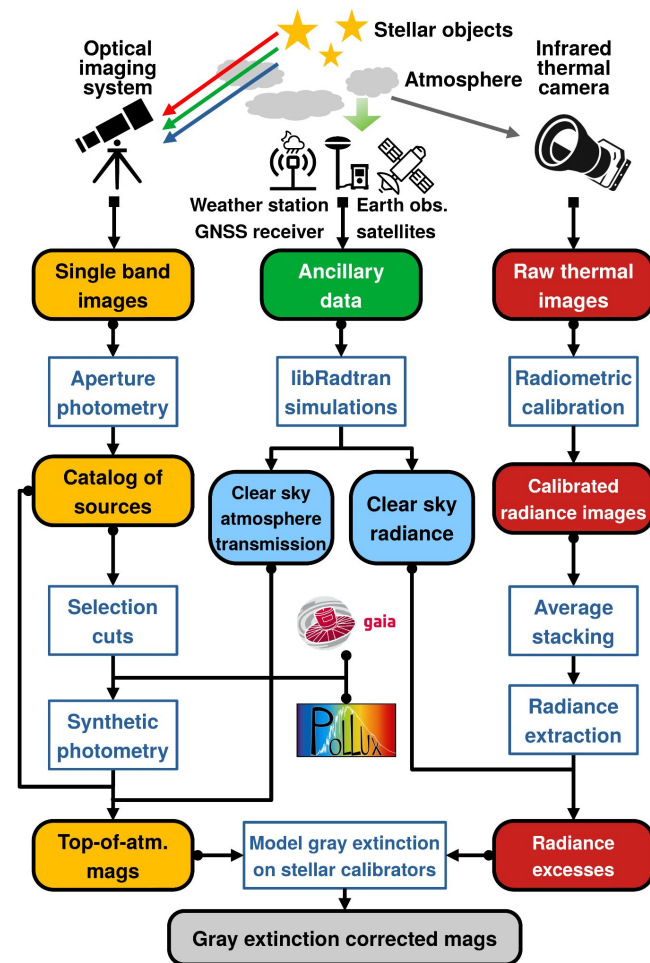


LWIR FOV



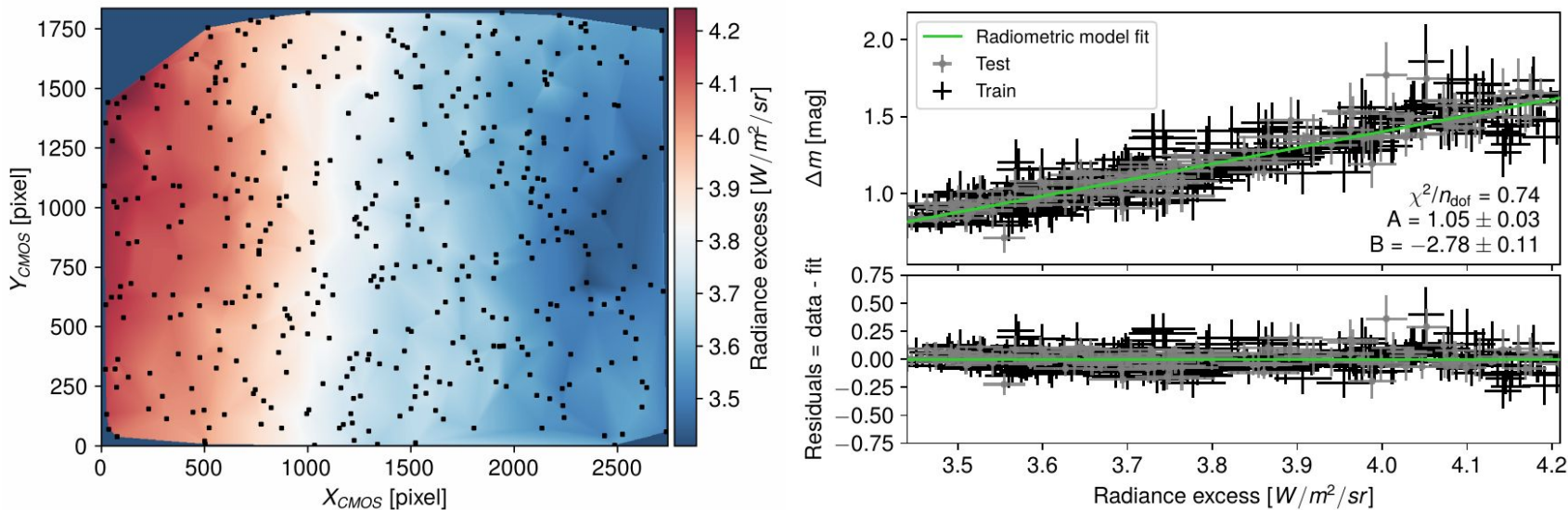
Data processing

1. For each observation epoch, a cloud-free atmosphere is simulated based on environmental conditions.
2. The radiance excess ΔL is computed as the difference between the total measured radiance at the position of a star in the radiometrically calibrated infrared image and the synthetic data.
3. Synthetic photometry is calculated using Gaia DR3 stellar parameters.
4. Chromatic atmosphere effects are removed from photometric observations.
5. Gray extinctions are computed for each star as the difference between true magnitudes and observed magnitudes.
6. Stars are separated into calibration and test groups, to emulate calibration star of known fluxes and transient objects respectively.
7. A model linking radiance residuals and magnitude differences (i.e., gray extinction) is fitted on the calibration group and applied to the test group to correct their observations.
8. Correction error is computed as the difference between the true magnitude and the corrected observed magnitudes.



Diagnostic plots after fit

Exposure n°2538 of the target field BD+28 4211.

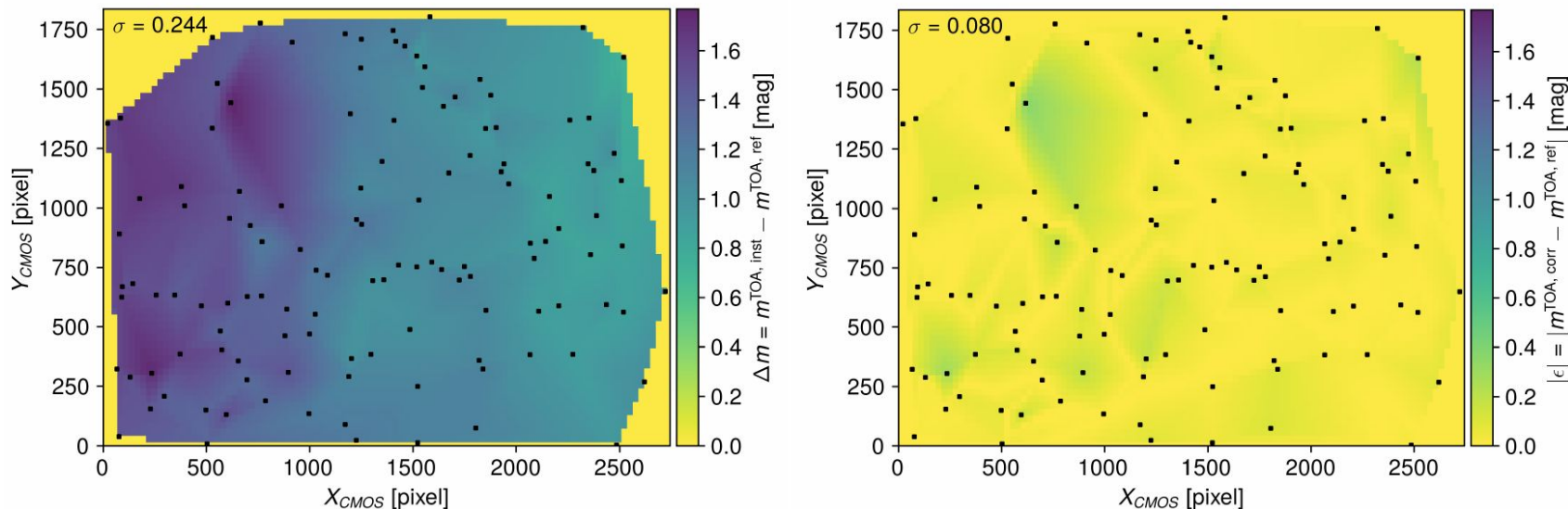


Left: Map of radiance excess projected onto the CMOS pixels coordinates grid. The radiance excess is shown for all stars (black dots) and linearly interpolated on a grid.

Right: Best-fit curve (solid red) of the gray extinction physical model versus radiance excess.

Diagnostic plots after fit

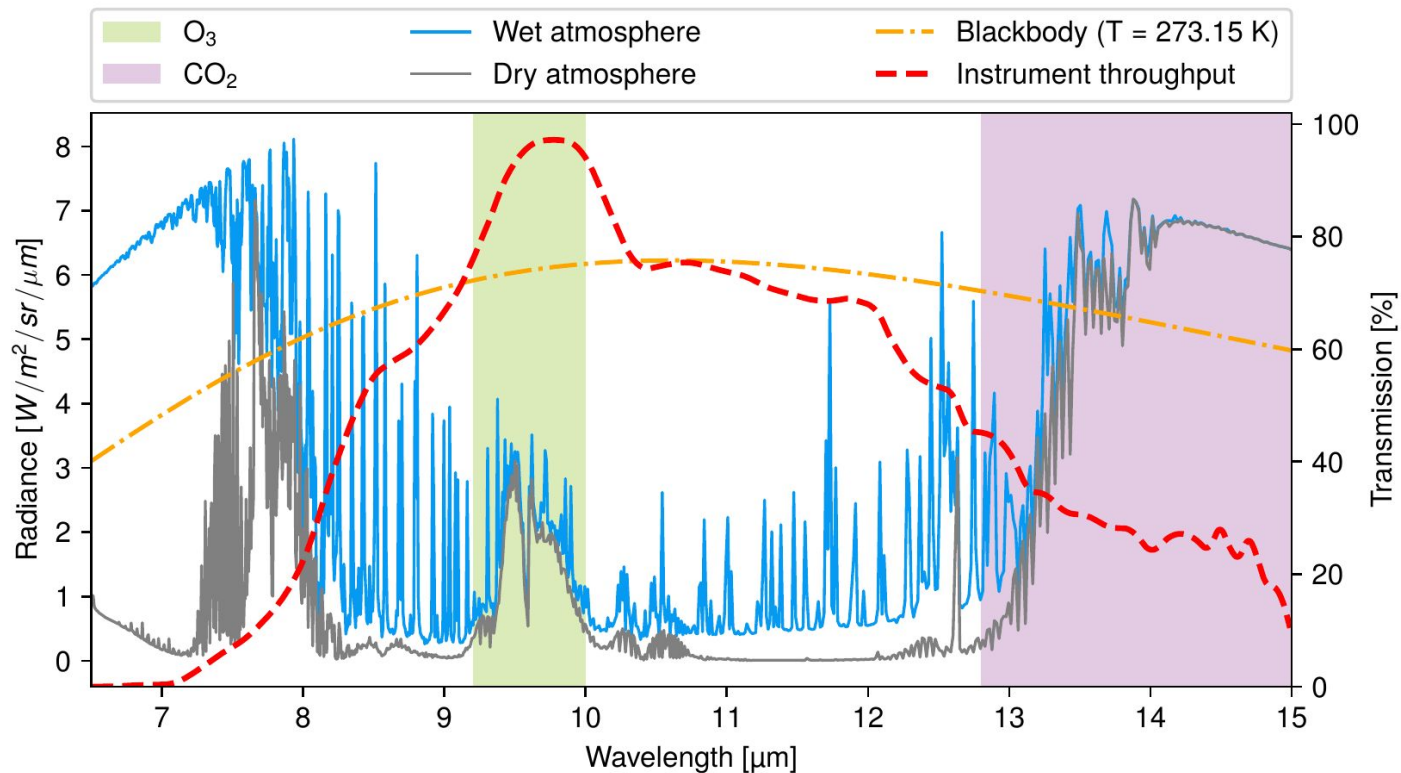
Exposure n°2538 of the target field BD+28 4211.



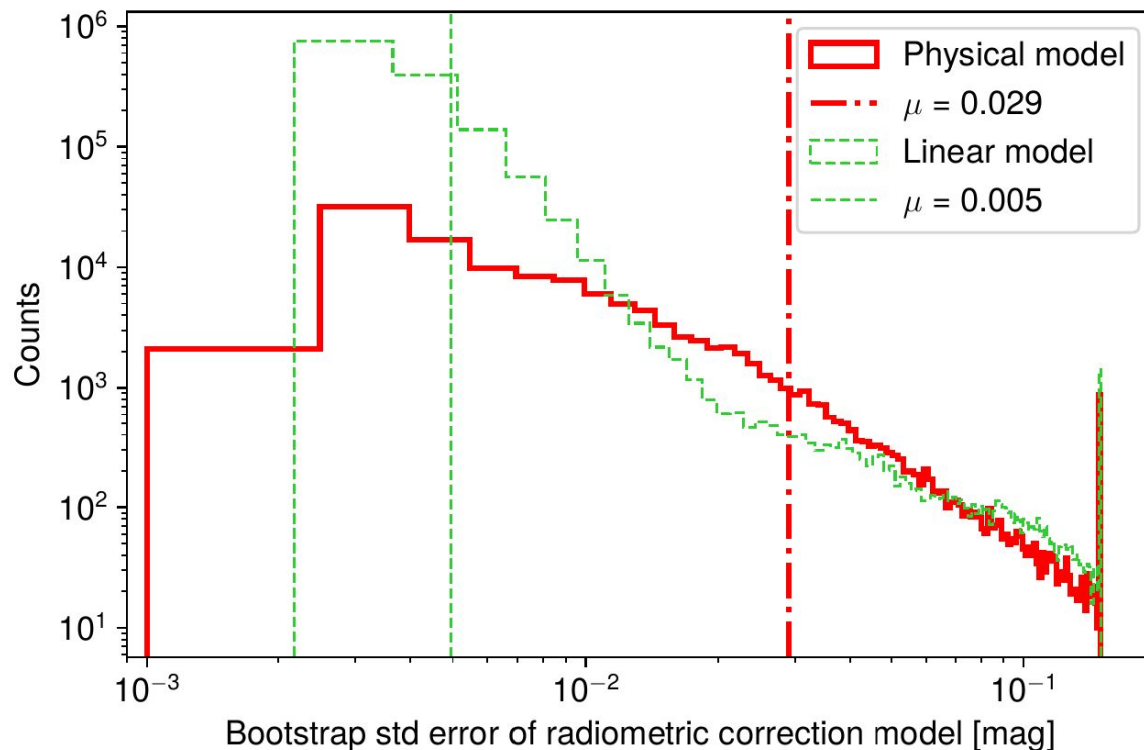
Left: Gray extinction before correction as a function of CMOS image coordinates. The extinction is shown for each test star (black dots) and linearly interpolated on a grid.

Right: Idem for the absolute value after correction. The colorbar range is kept identical to ease comparison.

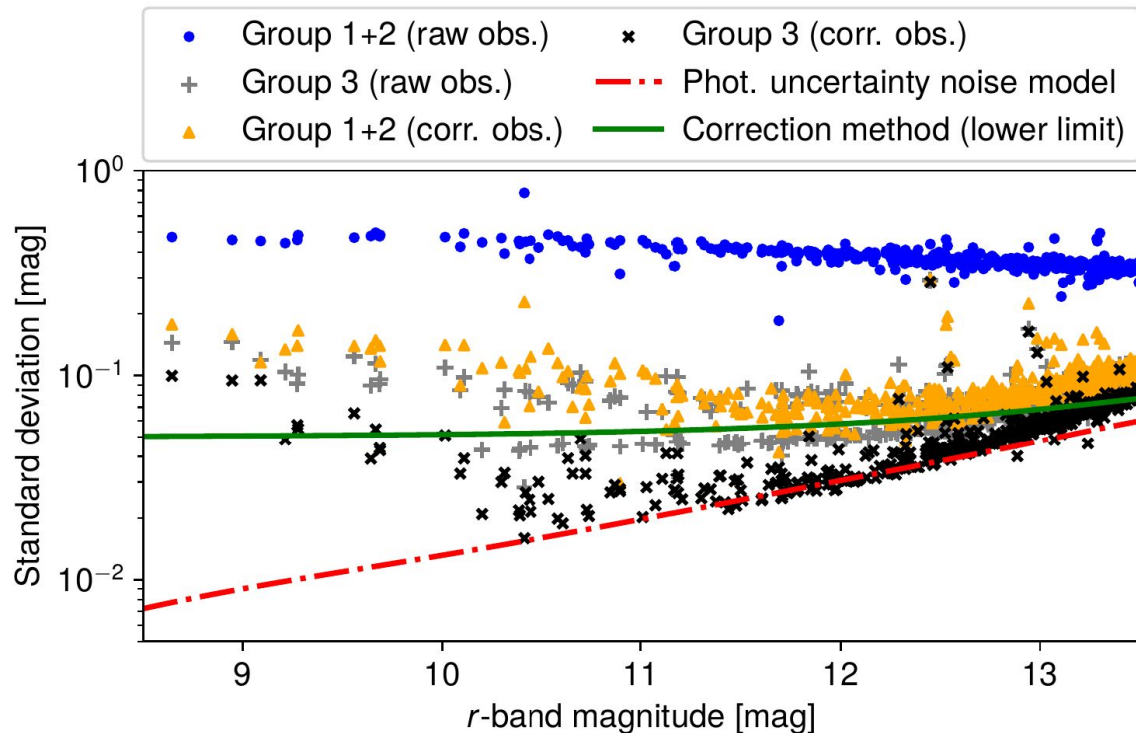
Atmosphere thermal emission in the LWIR band



Zero-point error via non-parametric bootstrap

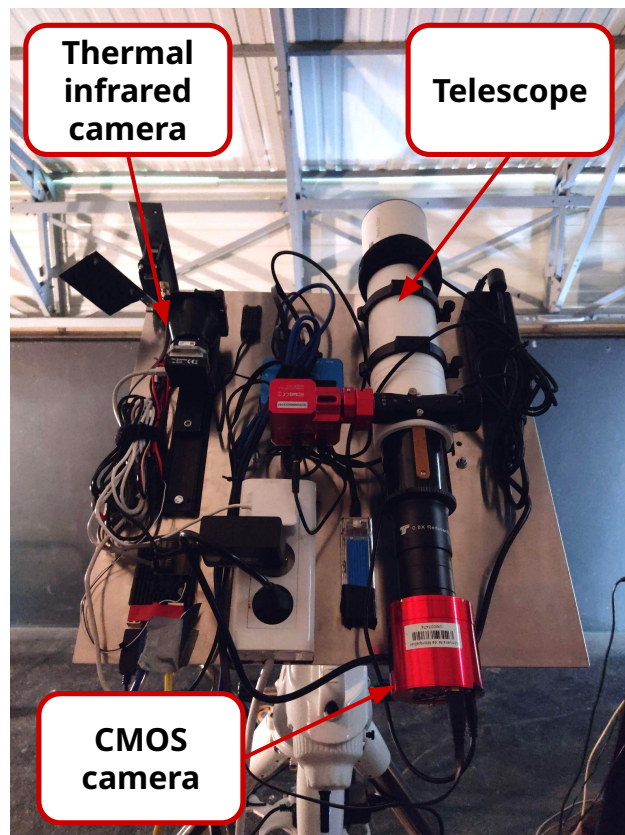


Results on all test stars



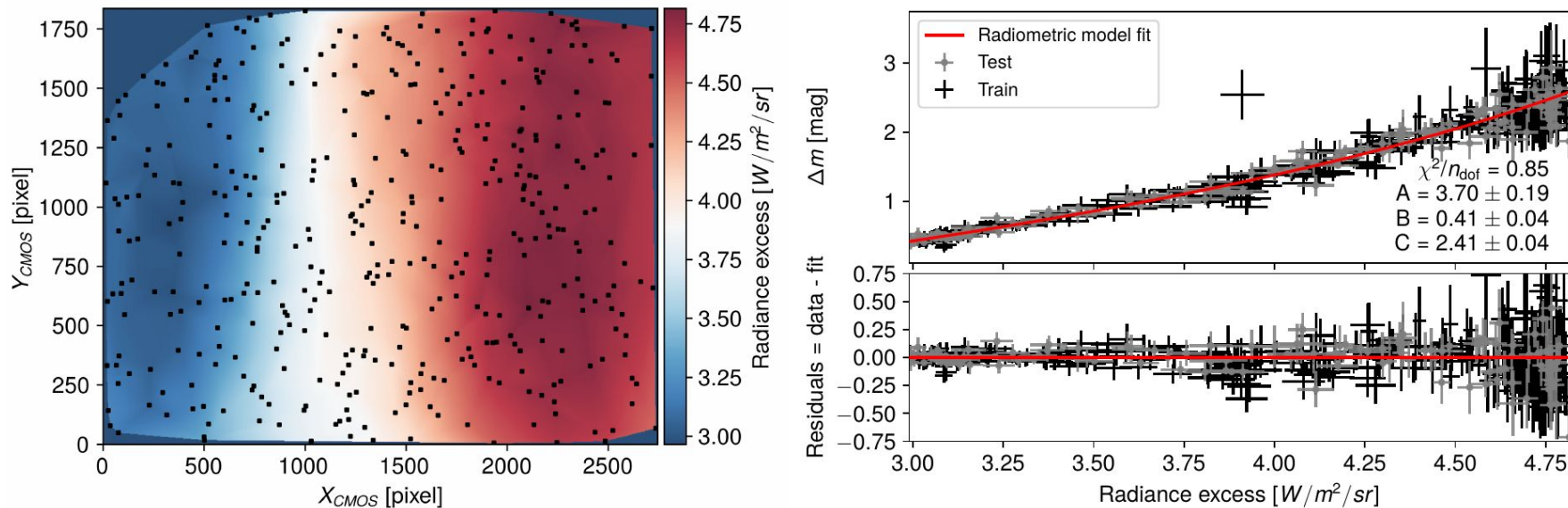
Experimental setup

- **Optical:**
 - 72-mm f/4.8 refractor
 - ZWO ASI 183 MM cooled CMOS camera
 - *r*-band filter
 - 2744×1836 pixels (bin 2×2)
 - $2.87''/\text{pixel}$
- **Infrared:**
 - FLIR Tau2 core
 - 60 mm f/1.25 athermalized lens
 - 640×512 pixels
 - $58.4''/\text{pixel}$
 - Calibrated in lab see Sommer et al., *Sensors* 2024, 24(14), 4498.
- **Atmosphere/environmental monitoring:**
 - Weather station
 - GNSS receiver



Diagnostic plots after fit

Exposure n°2548 of the target field BD+28 4211.

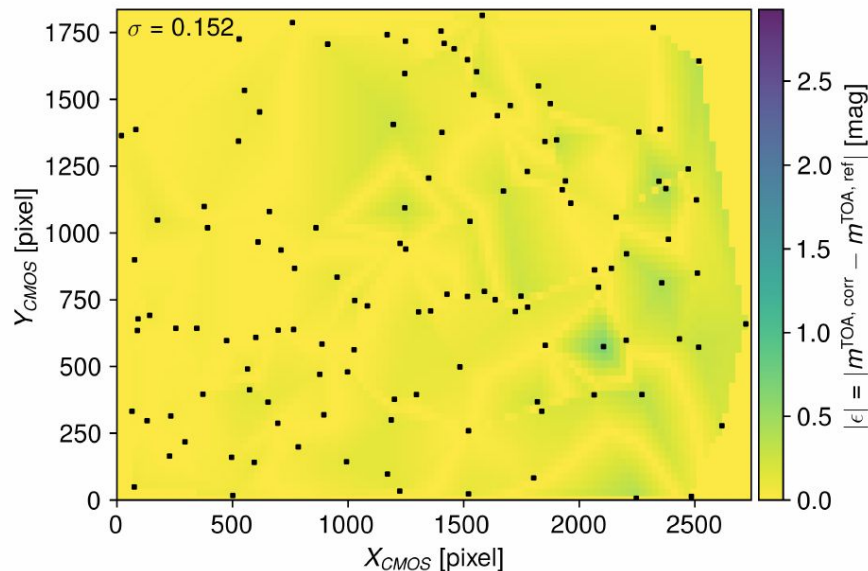
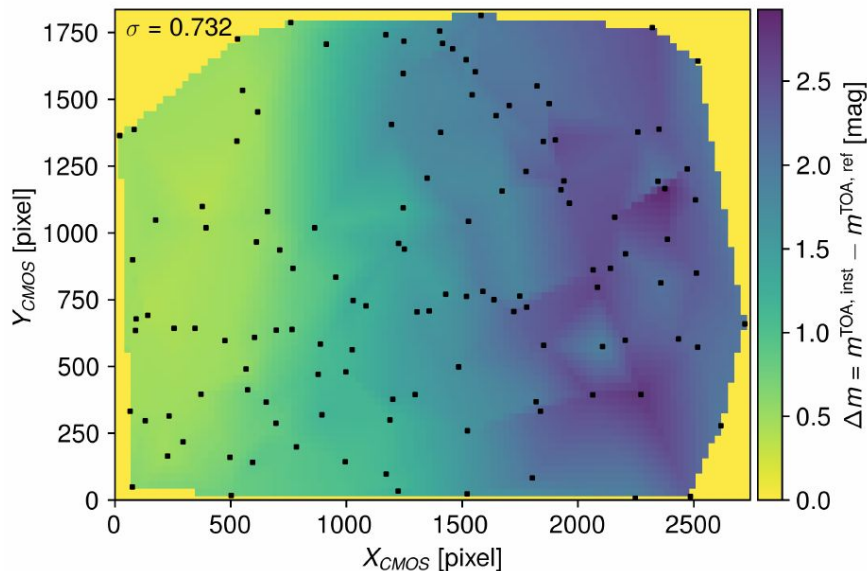


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Right: Best-fit curve (solid red) of the gray extinction physical model versus radiance excess.

Diagnostic plots after fit

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